

Marcelo del Valle Ph.D.

Research Assistant
NSF Center for Energy Smart Electronic Systems (ES2)
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Summary

Thermal engineer with experience in industry and academia. Strong background in experimental design and computational fluid dynamics (CFD). Excellent communication and leadership skills. Fluent in English and Spanish

Research Areas

Data centers thermal management and energy efficiency, computational fluid dynamics and heat transfer, thermal testing and component characterization

Education

- Present **Ph.D., Villanova University**, PA, expected July 2015. Dissertation topic: “Transient Numerical and Experimental Investigation of Data Center Hybrid Cooling Systems”
- December 2008 **M.S. in Mechanical Engineering, University of Nevada, Reno, NV**. Thesis title: “Benchmark and Sensitivity Study of the Container Analysis Fire Environment (CAFE-3D) Computer Code using a Rail-Cask-Size Pipe Calorimeter in Large-Scale Pool Fires”
- August 2000 **B.S. and professional license in Mechanical Engineering, Universidad de Santiago de Chile**, Chile. Thesis title: “Study and Analysis by Computational Simulations of the Fluid Flow and Heat Transfer in Channels with Geometrical Inhomogeneities” (Note: the B.S. degree in Chile is a 6 year program equivalent to a B.S./M.S. degree in the U.S.)

Professional Experience

- 08/09-Present **Research assistant:** Villanova University, Laboratory for Advanced Thermal and Fluid Systems and the NSF Center for Energy Efficient Systems, Projects: “Models and Metrics for Dynamic Air and Hybrid Liquid Cooled Data Centers Based on Computational and Experimental Approaches” and “Thermodynamic Tools for Holistic Analysis and Optimization of Energy Efficient Data Centers”. P.I. Dr. Alfonso Ortega
- 01/09-07/09 **External consultant for energy efficiency projects**, FEPASA railway, Chile.
- 06/06-12/08 **Research assistant:** University of Nevada, Reno, NV. Project: US Department of Energy, "AFCI Transportation Research". Research related to large-scale experiments and computational studies of heat transfer to massive objects engulfed in pool fires. P.I. Dr. Miles Greiner
- 08/04-03/06 **Department head for projects in mechanical engineering**, FEPASA railway, Chile.
- 01/03-07/04 **Thermal engineer:** JHG Engineering, Chile.
- 03/00-03/06 **Research assistant:** Universidad de Santiago de Chile, Projects: “Chaotic Advection and Flow Mixing in Flows Channels” and “Implementation and Technical Backup of Cluster for Parallel Processing”. P.I. Dr. Amador Guzmán

03/98-12/00

Research assistant: Universidad de Santiago de Chile, Projects: “Flow Mixing and Heat Transfer Enhancement in Laminar and Transitional Flows”, Fondecyt (Chile) research grant N° 1000003, 2000-2002 P.I Dr. Amador Guzmán

Technical Skills

Comsol, Ansys/Fluent, 6sigma room, FORTRAN, LabVIEW, Maple, MathCAD, Matlab/Simulink, Modelica

Publications

Journal Papers

1. Miles Greiner, Marcelo del Valle, Carlos Lopez and Victor Figueroa, 2013, “Thermal Measurements of a Rail-Cask-Size Pipe-Calorimeter in Jet Fuel Fires,” Journal of Fire Protection, Vol. 23, pp. 300-319
2. Amador M. Guzmán and Marcelo Del Valle, 2006, “Heat Transfer Enhancement in Grooved Channels due to Flow Bifurcations,” Journal of Heat and Mass Transfer (Wärme- und Stoffübertragung), Volume 42, Issue 11, pp. 967-975

Peer-reviewed conference papers

3. Marcelo del Valle and Alfonso Ortega, 2015 “Experimental Characterization of the Transient Response of Air/Water Cross Flow Heat Exchanger for Data center Cooling Systems” InterPACKICNMM2015, accepted to be presented at the ASME 2015 International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems and ASME 2015 13th International Conference on Nanochannels, Microchannels, and Minichannels
4. Tianyi Gao, Marcelo del Valle, Alfonso Ortega and Bahgat Sammakia, 2015 “Numerical and Experimental Characterization of the Transient Effectiveness of a Water to Air Heat Exchanger for Data Center Cooling Systems” InterPACKICNMM2015, accepted to be presented at the ASME 2015 International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems and ASME 2015 13th International Conference on Nanochannels, Microchannels, and Minichannels
5. Luis Silva-Llanca, Marcelo del Valle and Alfonso Ortega, 2015 “The effectiveness of Data Center Overhead Cooling in Steady and Transient Scenarios: Comparison of Downward Flow to a Cold Aisle Versus Upwards Flow From a Hot Aisle” InterPACKICNMM2015, accepted to be presented at the ASME 2015 International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems and ASME 2015 13th International Conference on Nanochannels, Microchannels, and Minichannels
6. Marcelo del Valle and Alfonso Ortega, 2014 “Numerical and Compact Models to Predict the Transient Behavior of Cross-flow Heat Exchangers in Data Center Applications” ITTherm2014, Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems
7. Luis Silva, Alfonso Ortega, Karman Fouladi, Marcelo del Valle and Vikneshan Sundaralingam, 2014 “Investigation of Exergy Destruction in CFD Modeling for a Legacy Air-Cooled Data Center” ITTherm2014, Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems
8. A.P. Wemhoff, M. del Valle, K. Abbasi and A. Ortega, 2013, “Thermodynamic Modeling of Data Center Cooling Systems” IPACK2013-73116 Proceedings of the ASME 2013 International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems

9. K. Abbasi, M. del Valle, A.P. Wemhoff and A. Ortega, 2013, "A Transient Model for Parallel Flow and Counter Flow Heat Exchangers" IPACK2013-73118 Proceedings of the ASME 2013 International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems
10. Devin Pellicone, Alfonso Ortega, Marcelo Del Valle, Steven Schon, 2011, "Simulation of Two-Phase Flow and Heat Transfer in Mini- and Micro-Channels for Concentrating Photovoltaics Cooling" ES2011-54206 Proceedings of the ASME 2011 5th International Conference on Energy Sustainability ES2011
11. Pellicone, D., Ortega, A., Delvalle, M., "Simulation of Two-Phase Flow and Heat Transfer for Practical Design of Mini- and Micro-Channel Heat Exchangers," Proceedings of the 2011 ASME International Mechanical Engineering Congress and Exposition
12. Greiner, M., del Valle, M., Lopez, C., Figueroa, V., and Abu-Irshaid, E., 2009, "Thermal Measurements of a Rail-Cask-Size Pipe-Calorimeter in Jet Fuel Fires," ASME 2009 Summer Heat Transfer Conference, HT2009-88520, July 19-23, 2009, San Francisco, California USA
13. Kramer, M.A., del Valle, M., and Greiner, M., 2008, "Measurement and Uncertainty of Heat Flux to a Rail-Cask Sized Pipe Calorimeter in a Pool Fire," PVP2008-61600, proceedings of the 2008 ASME Pressure Vessels and Piping Conference, PVP2008, July 27-31, 2008, Chicago, Illinois, USA Conference Paper
14. Del Valle, M.A., Kramer, M.A., Lopez, C., Suo-Anttila, A., and Greiner, M., 2007, "Temperature Response of a Rail-Cask-Size Pipe Calorimeter in Large-Scale Pool Fires", proceedings of the 15th International Symposium on the Packaging and Transportation of Radioactive Materials (PATRAM)
15. Amador M. Guzmán, Marcelo Del Valle, Pablo E. Araya and Alex M. Carrasco, "Flow bifurcations and heat transfer enhancement in channels with geometrical inhomogeneities," 2002 IMECE/ASME Conference, New Orleans, USA, November
16. Marcelo Del Valle, Alex M. Carrasco and Amador M. Guzmán, "Flow transitions and Heat transfer in Open Block Tandem Channels," ITherm 2002, Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems, May 29 - June 1, San Diego, California

References

Alfonso Ortega, Ph.D.

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